

EV Charge and Range Visualization - Starter Code

Save the following Python code as **ev_visualization.py** and run it after installing pandas, numpy, matplotlib and seaborn.

```
import os, pandas as pd, numpy as np, matplotlib.pyplot as plt
import seaborn as sns
from mpl_toolkits.mplot3d import Axes3D
os.makedirs("charts", exist_ok=True)
np.random.seed(1)
df = pd.DataFrame({
    "Vehicle": [f"EV{i}" for i in range(1,31)],
    "Battery_kWh": np.random.randint(30,100,30),
    "Range_km": np.random.randint(180,600,30),
    "Charge_Time_hr": np.random.uniform(0.5,8,30),
    "Efficiency": np.random.uniform(4,9,30),
    "Type": np.random.choice(["SUV", "Sedan", "Hatchback"], 30)
})
df.to_csv("dataset.csv", index=False)
df.plot.bar(x="Vehicle", y="Range_km")
plt.tight_layout()
plt.savefig("charts/bar.png")
plt.close()
plt.plot(df["Vehicle"], df["Range_km"])
plt.xticks(rotation=90)
plt.tight_layout()
plt.savefig("charts/line.png")
plt.close()
df["Type"].value_counts().plot.pie(autopct="%1.1f%%")
plt.ylabel("")
plt.savefig("charts/pie.png")
plt.close()
plt.scatter(df["Battery_kWh"], df["Range_km"])
plt.savefig("charts/scatter.png")
plt.close()
sns.histplot(df["Range_km"])
plt.savefig("charts/histogram.png")
plt.close()
sns.boxplot(y=df["Charge_Time_hr"])
plt.savefig("charts/boxplot.png")
plt.close()
sns.violinplot(y=df["Efficiency"])
plt.savefig("charts/violin.png")
plt.close()
sns.heatmap(df.select_dtypes("number").corr(), annot=True)
plt.savefig("charts/heatmap.png")
plt.close()
fig=plt.figure()
ax=fig.add_subplot(111, projection="3d")
ax.scatter(df["Battery_kWh"], df["Charge_Time_hr"], df["Range_km"])
plt.savefig("charts/scatter3d.png")
plt.close()
print("Done")
```